AVROD — Project Report

Automated Verification for Restricted Operator Descent (formerly AVRODH) · VIVO Ignite National Top 10

1. Executive Summary

AVROD is a mechanization-first manhole safety system that manages access and hazard alerts, while its M-PACT interface allows different cleaning tools to connect to the same manhole. It functions as an IoT-based safety interlock for sewer and septic manhole entry, built to physically prevent unsafe manual descent by sanitation workers, and also provides a universal interface for easy attachment of various cleaning machines.

The project began as **AVRODH** (Automated Verification for Restricted Operator Descent into Hazard), was shortlisted in VIVO Ignite's Top 200, Top 30, and now the **National Top 10**, and has evolved through mentor feedback into a broader framing that places mechanized cleaning first and treats authorized manual entry as a safety-enforced exception rather than the default approach.

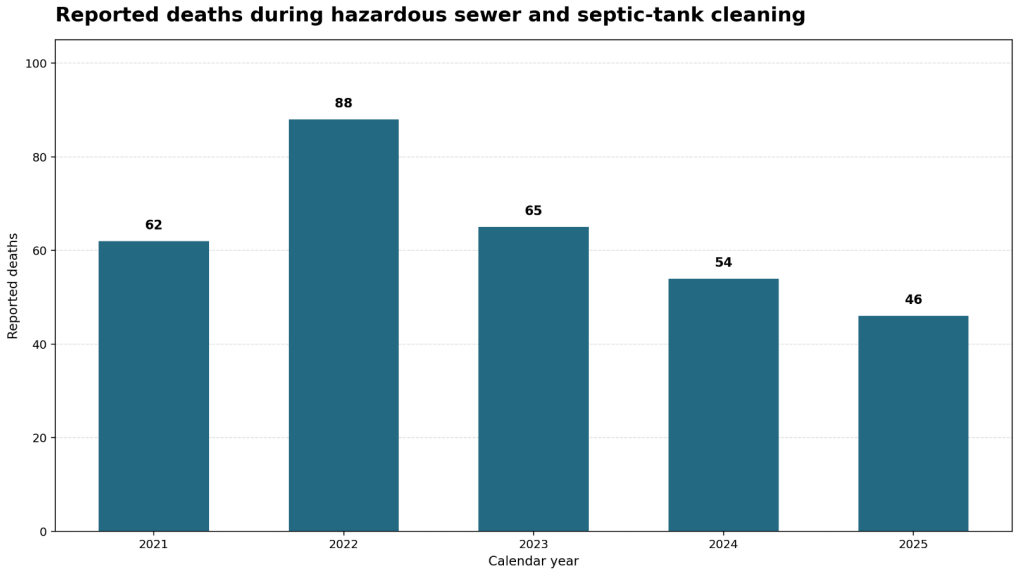
This report documents the problem the project addresses, how the system is designed to work, how the idea evolved under mentorship, and a clear account of what has been built and tested so far versus what remains proposed or in progress.

2. The Problem: India's Sewer Safety Crisis

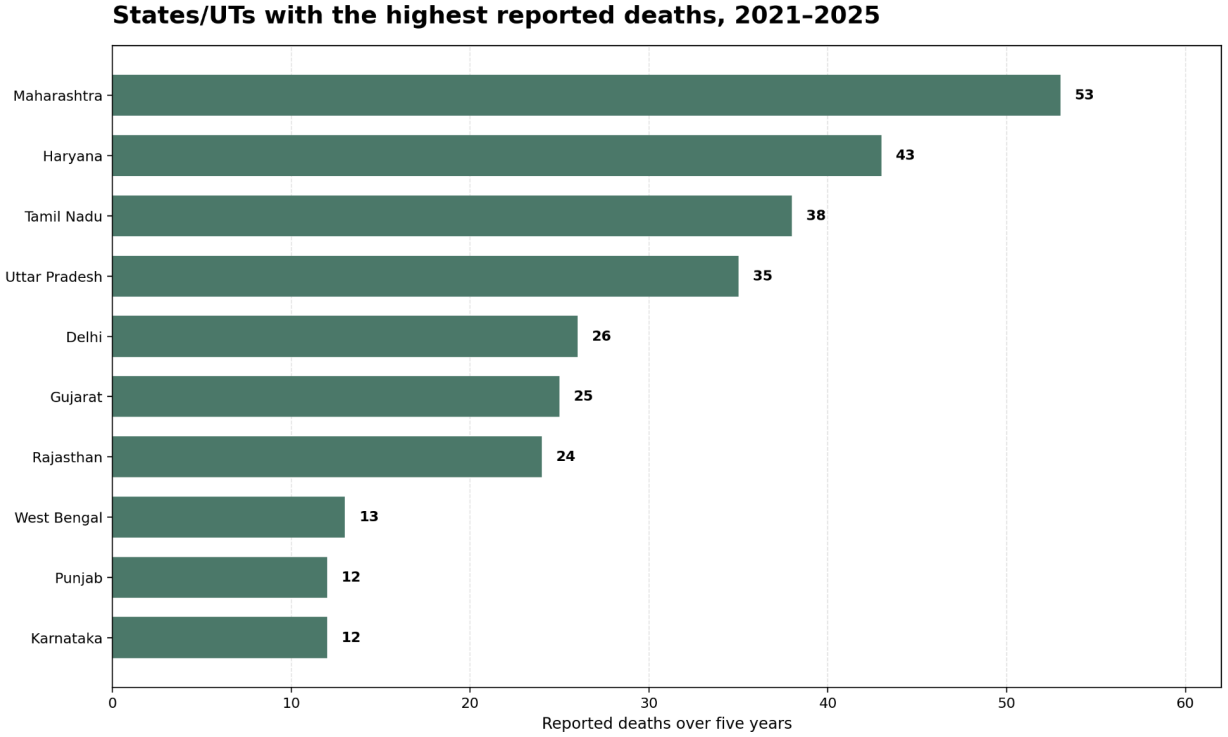
Between 2014 and 2025, **859 sanitation workers died** cleaning Indian sewers and septic tanks — an average of over 70 deaths a year, according to National Commission for Safai Karamcharis (NCSK) data.

Despite public assurances that “technology now handles such inhumane work,” manual entry remains widespread in low-resource municipalities despite the legal ban on manual scavenging, and mechanization is still incomplete nationwide. Gas detectors and PPE often exist on paper but go unused in practice. The core failure is not an absence of technology — it is an absence of enforcement.

Source: National Commission for Safai Karamcharis (NCSK) data, as analyzed in “Sewer Worker Deaths Continue Despite Ban and Safety Policies,” Dataful, citing figures as on 31 December 2025.



Source: Rajya Sabha Unstarred Question No. 3023, answer dated 18 Mar 2026, annexure yearly totals. Note: these yearly figures sum to 315; a later reply dated 1 Apr 2026 reports 317 deaths for 2021–2025. Resolve this official-data discrepancy before final publication.



Source: Rajya Sabha Unstarred Question No. 3023, answer dated 18 Mar 2026, state-wise annexure. This chart shows the 10 largest state/UT totals in that annexure; it is not a risk rate adjusted for population or workforce.

3. The Mundka Tragedy — 26 June 2026

The emotional and practical starting point for this project was a tragedy in the Mundka Industrial Area of Outer Delhi. Three daily-wage laborers — Arun (38), Sandeep (32), and Chand (42) — were hired by a local contractor for ₹20,000 to clear a septic tank at a commercial printing unit that hadn't been cleaned in over a year.

The workers initially used motorized suction to pump out wastewater, but once the liquid was gone, the contractor withheld payment unless they entered manually to clear the remaining toxic sludge by hand, for a promised ₹3,000 each. The tank held toxic gas that had built up over months, compounded by chemical effluent from the press. The first worker was overcome almost immediately; the second and third, entering one after another to rescue their colleagues, were overcome within seconds. All three died. Delhi Fire Service trucks took over 40 minutes to reach the site through industrial traffic, and firefighters could not safely enter until equipped with breathing apparatus.

This incident is the direct motivation behind AVROD: a system that makes unsafe entry physically impossible, rather than relying on guidelines that go unenforced.



4. Solution Overview: How AVROD Works

AVROD is a rugged, electronically locked hub fitted to a manhole or sewage entry point. It unlocks only when three independent conditions are verified:

- **Gas sensor check** — confirms no hazardous gas levels are present inside.
- **RFID verification** — RFID tags on PPE components confirm the worker is geared up before entry.
- **Municipal authorization** — a code sent to the municipal authority must be entered, so every entry is accountable to a named authority, not just the worker or contractor.

The same hub also carries M-PACT (Modular Port for Attaching Cleaning Tools) — the standardized manhole-side interface detailed in Section 10. This means the installed site isn't limited to the entry interlock alone: it supports mechanized cleaning equipment connecting at the same manhole as the first-
choice approach, with the gas/RFID/authorization interlock described above enforcing safety for the
exception case where mechanized cleaning isn't yet available and authorized manual entry is required

Any attempt made without completing all three conditions is denied, and the system is designed to alert the relevant authority and log the attempt with a date-time stamp — intended to create an auditable safety-compliance record over time. The device's physical schema includes a hinged hub (flips up in use, flattens flush when idle), a keypad and display for code entry, and a gas sensor suspended inside the shaft. The proposed design also includes post-entry monitoring — continuous gas tracking and a worker-motion "dead man's switch" via a harness-mounted sensor — with auto-retraction through a winch and cable-hook if conditions turn unsafe or the worker stops moving.

Note on scope: Section 15 (Limitations & Honest Scope) gives the current, confirmed status of every subsystem — what's tested and working versus what remains a designed, proposed capability.

NOTE - Our first priority is always to ensure that no human has to enter for cleaning purposes. That is the sole reason behind why I added M-PACT to AVROD. But during some circumstances like routine repair and checkup, and some places where machines fail, human entry is unavoidable and hence it is also mentioned in the cleaning SOP published by the government (refer). Therefore my project doesn't promote human entry but ensures reinforcement of law.

5. Scientific Approach

AVROD applies **interlock logic** — the same principle behind industrial machine lockout systems, where multiple independent safety conditions must all be verified before a hazardous action is physically permitted. The design combines a gas-threshold sensor, RFID presence detection, and accountable authorization into a single safety mechanism, with provision for both direct electrical

supply and solar power depending on site conditions. It also features a dead man's switch in the form of worker's motion and vitals. If any anomaly in worker's vitals is detected, which is constantly metered by wearables that are a part of the PPE kit (along with RFID tags for each component), the worker is immediately pulled back via a winch. This automatic system is crucial because, when one suffocates in the depths, he doesn't realise, his body just slowly shuts down even without any warnings.

6. Flow of control

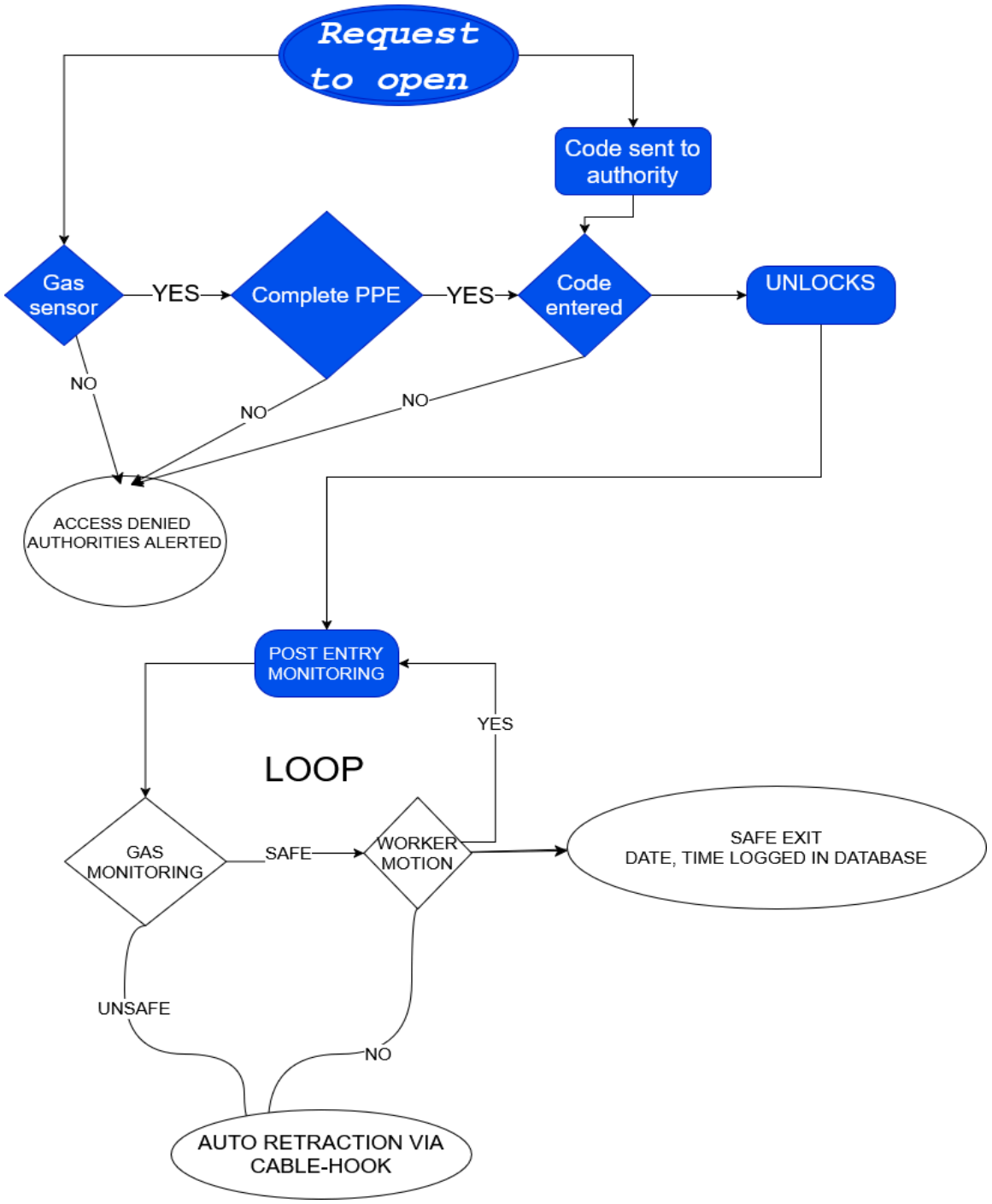


Figure 1. End-to-end interlock and post-entry monitoring logic.

7.1 Schema (v1)

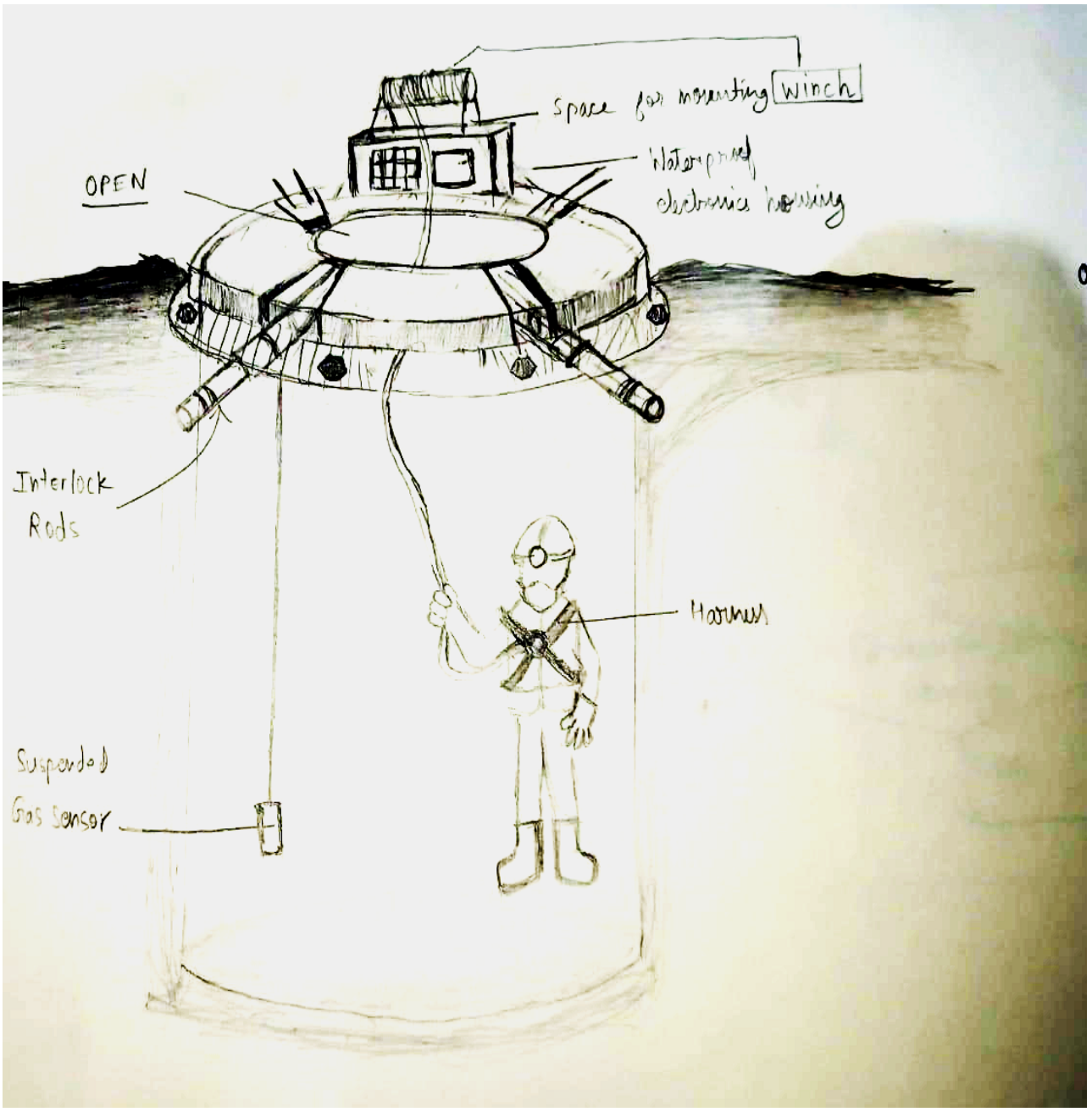


Figure 2. Original device schema — hinged hub, interlock rods, suspended gas sensor, and winch mounting space.

7.2 Schema (v2)

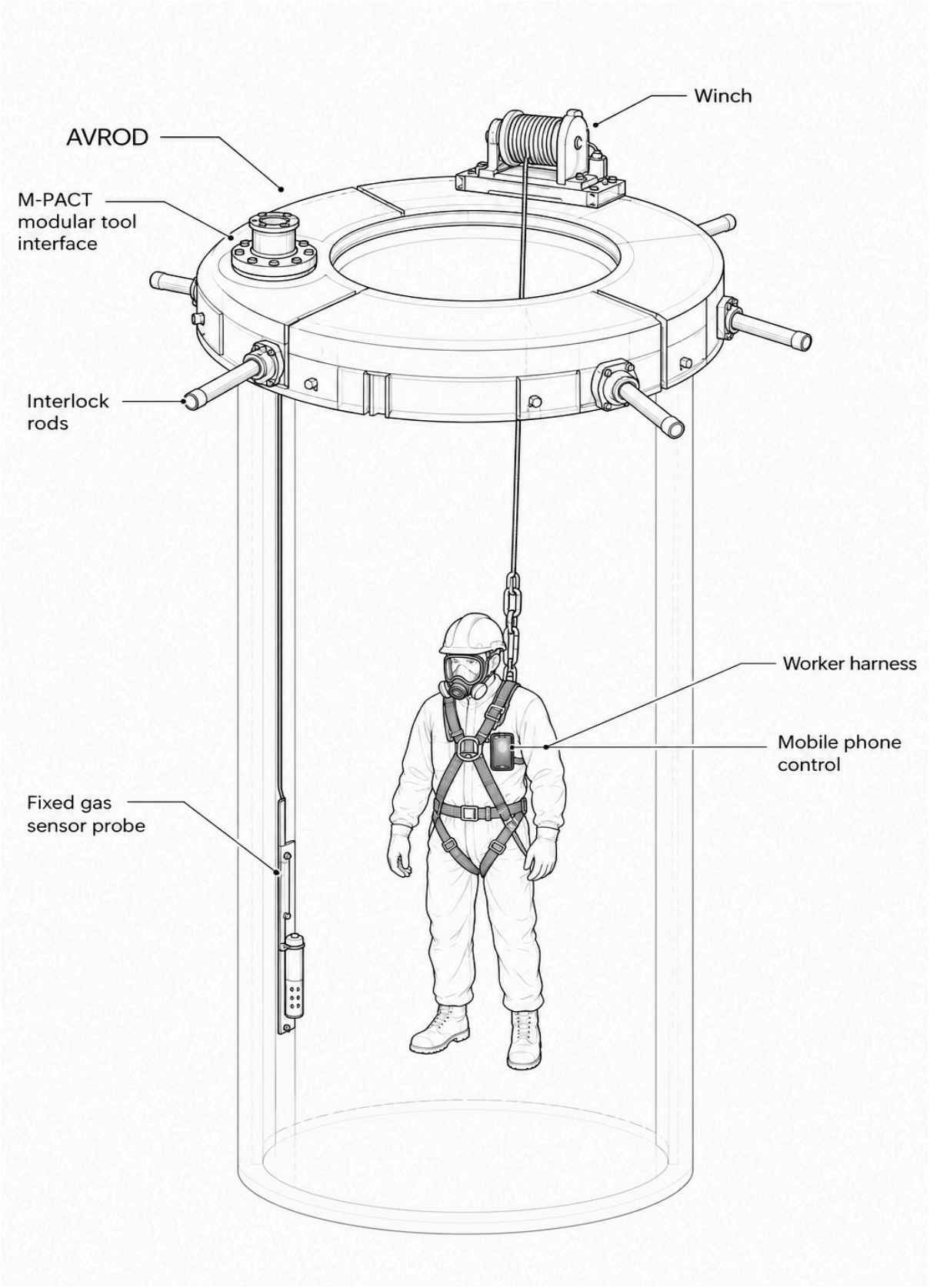


Figure 3. New refined device schema — M-PACT and winch mounting space.

8. Objective

The objective is to make unsafe sewer entry physically impossible in municipalities that still lack full mechanization — not to make manual entry acceptable, but to ensure that where it happens, it happens only under verified, enforced safety conditions, with continued protection during the task itself.

9. Modular Port for Attaching Cleaning Tools (M-PACT)

M-PACT was conceived as a shared physical interface layer at the manhole mouth — infrastructure that different mechanized cleaning methods (suction, jetting, rodding, bucket/clamshell) could use in common, rather than each requiring custom site adaptation. Combined with AVROD's safety-enforcement layer and a longer-term data/AI layer for gas-pattern mapping across a sensor network, the concept positions mechanization as the first-choice approach, with AVROD's interlock serving authorized, exceptional manual entry rather than routine use.

The sensors will provide us with enough data to train models and create heatmaps to pinpoint the exact causes of toxic affluent buildup in sewages.

10. Estimated Cost

Material	Estimated Cost (INR)
Microcontroller (ESP32)	350
Gas sensor module	200
RFID tag + reader	150
Keypad	150
LCD screen	150
Metal interlock rod + actuator	2,500
Hinge mechanism	300
Power supply and wiring	400
Casing / housing	300
Miscellaneous	300
Total	**₹4,800**

11. Commercial Viability & Scalability

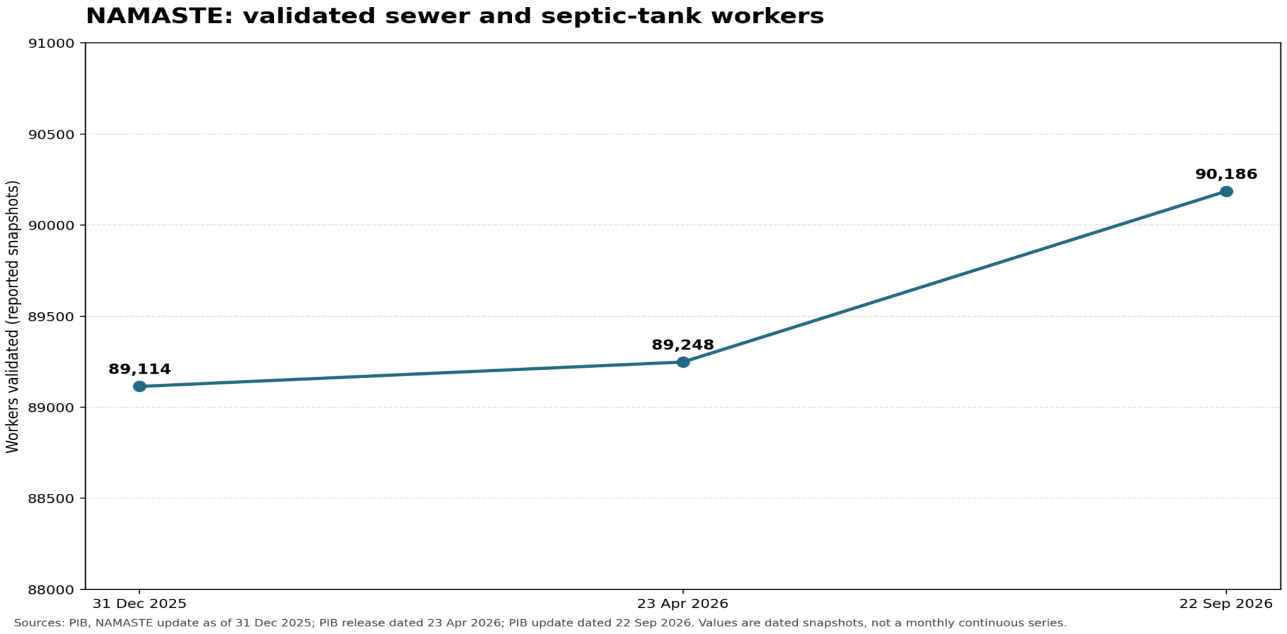
At an estimated per-unit cost of roughly ₹4,000–4,900 at streamlined production, AVROD is deployable at a scale and price point far below robotic alternatives — publicly reported Bandicoot robot procurements range from about ₹4 lakh to **₹32 lakh** (Greater Hyderabad Municipal Corporation's CSR-funded procurement), positioning AVROD as a complementary safety layer for low-resource municipalities rather than a competing replacement for full mechanization.

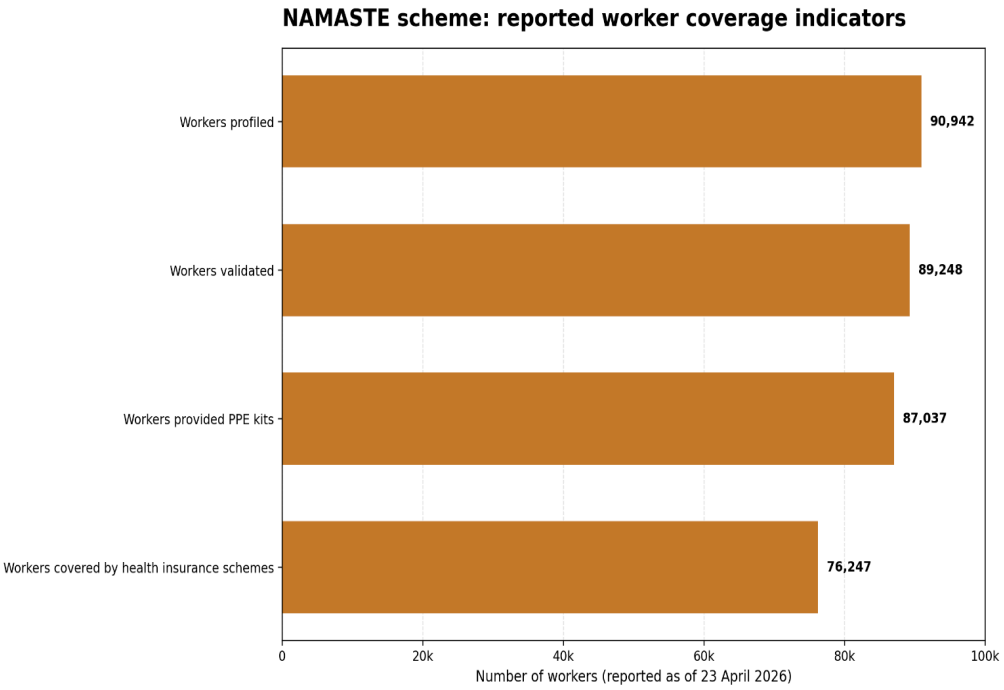
12. Target Population

Sanitation workers in Indian municipalities that still rely on manual sewer and septic entry, particularly low-resource municipalities where full mechanization has not yet been deployed.

13. Social Impact

The workforce performing manual sewer and septic cleaning in India is drawn overwhelmingly from marginalized castes. AVROD is intended to address both an acute physical safety crisis and a long-standing social justice issue in Indian sanitation labor, by making unsafe entry structurally harder rather than relying solely on policy or enforcement that has historically gone unapplied.





Source: Ministry of Social Justice & Empowerment / PIB, "Empowering Lives: NAMASTE Scheme Delivers Tangible Change for Sanitation Workers", 23 Apr 2026. These are different program indicators, not mutually exclusive groups and not evidence that AVROD itself prevents deaths.

14. CAVEATS & SOLUTIONS

- The project acted as a low-cost safety layer — described by the previous submissions as a "seatbelt" — until M-PACT was added. It now doesn't act like merely a seatbelt, it is an active part of the manhole system that has its own role. It can be described as a USB C interface for manholes. All the new additions have now made it capable of easy connection with common cleaning methods.
- Each installation requires site-level installation and maintenance; it is not a zero-maintenance solution. This can only be tackled post installation which will provide us data on most common maintenance requirements and hence I can innovate a solution accordingly
- The physical MQ2 gas sensor is tuned to flag anomalous readings, and the interlock's gas-hazard logic is validated through a controllable multi-gas simulator (H_2S, CH_4, CO).
  A field deployment would pair this with a calibrated, hazard-appropriate sensor (e.g. MQ136 for H_2S).
- The servo-based interlock now holds firmly in testing. As with any mechanical lock, long-term durability under sustained field use will continue to be validated as deployment scales.
- RFID verification confirms possession of an authorized token, not that PPE is correctly worn. Hence at next stage we aim to add a camera feature to the web interface that checks that PPE is worn correctly.
- M-PACT and the future AI/data-pattern layer are architectural concepts, not yet implemented or field-tested systems.

15. Bibliography

- National Commission for Safai Karamcharis (NCSK), 2024; Ministry of Social Justice & Empowerment, Parliamentary reports.
- Parliamentary Standing Committee on Urban Development, 2021.
- “Sewer Worker Deaths Continue Despite Ban and Safety Policies,” Dataful (insights.dataful.in), citing NCSK data as on 31 December 2025.
- Greater Hyderabad Municipal Corporation Bandicoot robot procurement cost, ETV Bharat, 2019.
- Central Public Health and Environmental Engineering Organization (CPHEEO); Ministry of Housing and Urban Affairs; Government of India; www.amrut.gov.in; November 2018.

16. Project Evolution — From AVRODH to AVROD

The project has gone through several real inflection points rather than following a straight line from idea to finished device:

- **Late June 2026** — The idea took shape around the Mundka tragedy, under deadline pressure for the VIVO Ignite submission.
- **24 July 2026** — Selected for the VIVO Ignite Top 200.
- **July–August 2026** — First physical prototype: a two-colour PLA model designed in Tinkercad, built around a NodeMCU/ESP8266, RFID reader, buzzer, and gas sensor. This first version did not include a physical locking mechanism. A project video, opening with the Mundka story, was submitted on 21 August 2026.
- **8 September 2026** — Selected for the National Top 30.
- **15 September 2026** — At the first Achiever 30 mentoring session, the mentor raised a serious concern: the framing could be read as supporting an approach inconsistent with the law and the intended direction of safe sanitation. This triggered a substantial rethink of the project's framing rather than an abandonment of it.
- **16 September 2026** — Work began on the Modular Port for Attaching Cleaning Tools (M-PACT) concept, under the guiding analogy of “USB-C for cleaning devices”: a standardized manhole-side interface so different mechanized cleaning systems could connect to shared infrastructure.
- **19 September 2026** — At the second mentoring session, the reframing was recognized. The mentor also suggested dropping the “H” from AVRODH, since “hazard” gave the name a negative framing. The project was renamed AVROD.
- **5 October 2026** — AVROD reached the National Top 10.
- **8–10 October 2026** — A new physical prototype was built directly in transparent acrylic, chosen for its clean, see-through look that lets the internal mechanism be seen during demonstrations.

17. FUTURE SCOPE

- ☐ Designing a rugged and waterproof outer case.
- ☐ First and foremost will be partnering with multiple municipalities to first realize the problems and innovate their solutions.
- ☐ Camera input on webpage to verify PPE kit is properly worn.
- ☐ Data accumulation of gas and toxic affluents buildup.
- ☐ Data will allow for anomaly identification and mapping hotpots.
- ☐ Installing sensors for acoustic sensing of structural integrity of sewages.
- ☐ Lite variant purely for data accumulation
- ☐ Designing solar based modules which will be plug and play power sources, compatible with all AVROD units.
- ☐ Connecting RFID PPE kits to NAMASTE Workers' Database

18. Milestones

Date	Milestone
~26–27 June 2026	Idea conceived following the Mundka tragedy; AVRODH submitted
24 July 2026	VIVO Ignite Top 200
July–August 2026	First physical prototype (no lock); project video submitted 21 August
8 September 2026	National Top 30
15 September 2026	Mentor feedback on framing
16 September 2026	M-PACT concept begins
19 September 2026	Reframing recognized; renamed AVROD
5 October 2026	National Top 10
8–10 October 2026	Acrylic prototype built